LMRA Problem – 3 **Date : 27-03-2026**

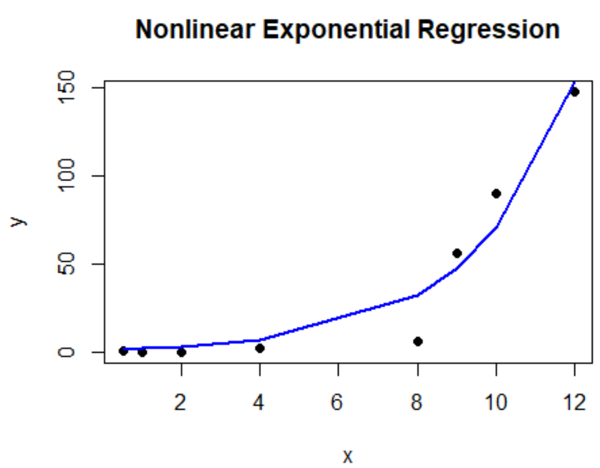
Consider the following observations x and y and fit the non-linear regression model y=theta1*e^theta2x+E to the given data.

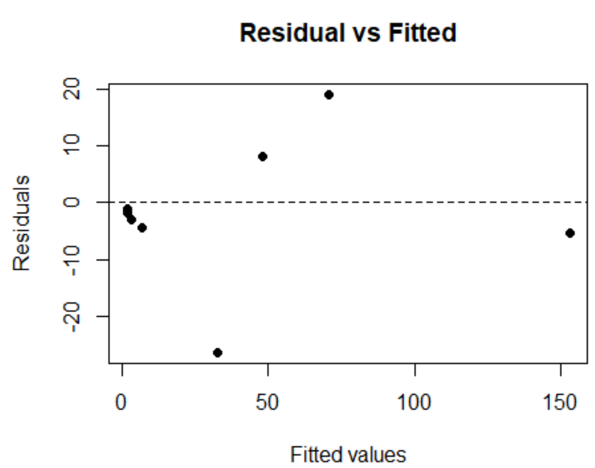
X	Y
0.5	0.68
1	0.45
2	0.25
4	2.5
8	6.19
9	56.1
10	89.8
11	147.7

Solution :-

```
# Data
x <- c(0.5,1,2,4,8,9,10,12)
y <- c(0.68,0.45,0.25,2.5,6.19,56.1,89.8,147.7)

data <- data.frame(x,y)

# Nonlinear exponential regression model
model <- nls(y ~ theta1 * exp(theta2 * x),
             data = data,
             start = list(theta1 = 1, theta2 = 0.4))

summary(model)

##
## Formula: y ~ theta1 * exp(theta2 * x)
##
## Parameters:
##        Estimate Std. Error t value Pr(>|t|)
## theta1  1.48397    1.08568   1.367 0.220670
## theta2  0.38637    0.06427   6.012 0.000955 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.09 on 6 degrees of freedom
##
## Number of iterations to convergence: 7
## Achieved convergence tolerance: 2.428e-06

# Predicted values
y_pred <- predict(model)

# Residuals
res <- y - y_pred

# Residual standard deviation
```

```
s <- sqrt(sum(res^2) / (length(y) - 2))
s

## [1] 14.09426

# R-square
SSE <- sum(res^2)
SST <- sum((y - mean(y))^2)
R2 <- 1 - (SSE / SST)
R2

## [1] 0.9446788

# Sort values for smooth curve
ord <- order(x)
x_sorted <- x[ord]
y_pred_sorted <- y_pred[ord]

# Plot data + fitted curve
plot(x, y, pch = 16,
     main = "Nonlinear Exponential Regression",
     xlab = "x", ylab = "y")

lines(x_sorted, y_pred_sorted, col = "blue", lwd = 2)


# Residual plot
plot(y_pred, res,
     main = "Residual vs Fitted",
     xlab = "Fitted values",
     ylab = "Residuals",
     pch = 16)

abline(h = 0, lty = 2)
```





Interpretation :

- The exponential model fits the data well because the predicted curve follows the increasing pattern of the points and the R^2 value is high.
- The residual plot shows no clear pattern, so the model assumptions look reasonable and the model is suitable for prediction.